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Hemoglobin Estimation from Smartphone-Based Photoplethysmography with Small Data

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***Abstract*—Photoplethysmography (PPG) is a well-known technique to estimate blood pressure, oxygen saturation, and heart frequency. Recent efforts aim to obtain PPG from wearable and mobile devices, allowing more democratic access. This paper explores the potential of using a smartphone camera as a PPG sensor, getting a time series based on the RGB values of video recordings of patients' fingertips. Through this PPG, we apply machine learning for the non-invasive estimation of hemoglobin levels. We assume a realistic scenario where the data has a low volume and potentially a low quality. The generalization capacity of the models built on these scenarios usually achieves undesirable performance. This paper presents a novel dataset that comprises real-world mobile phone-based PPG and a comprehensive experiment on how different techniques may improve hemoglobin estimation using deep neural architectures. In general, cleaning, augmentation, and ensemble positively affect the results. In some cases, these techniques reduced the mean absolute error by more than thirty percent.**

***Index Terms*—mHealth, deep learning, time series, hemoglobin, photoplethysmography**

I. Introduction

Measuring clinical parameters is an essential step for diagnosing and treatment decisions regarding many diseases. Unfortunately, the most common approaches to obtaining these parameters rely on invasive methods, using medical tools such as needles and syringes. Even for fluids that can be collected without invasive methods, such as urine, the transportation and storage processes can impact the result of the samples analysis [1], [2]. Alongside these problems, many clinical parameter assessments require high-cost equipment for analysis and highly-qualified employees to operate them.

As a way to overcome these issues, many non-invasive techniques were proposed to obtain physiological data precisely. However, these techniques still rely on expensive or heavy equipment and, consequently, are usually restricted to hospitals or a small part of the population. These difficulties have motivated the creation of electronic and mobile health (eHealth and mHealth, respectively) techniques. Due to these factors, new techniques and devices for non-invasive, low-cost diagnosis and treatment of medical conditions are increasingly emerging [3].

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Furthermore, rapid tests for diagnosis, known as *point-of-care testing* (POCT), emerged from the necessity of turning diagnosis available to a great variety of places, especially those situated in remote localizations and with few technology [4]. Recently, the use of smartphones for the collection and sending of that has become more relevant in the context of telemedicine, since it's a non-invasive way of obtaining useful information for the performing of POCT with a wide range of purposes [5], [6].

This work explores cellphone-based photoplethysmography (PPG) to estimate hemoglobin levels. Because of its simplicity and utility, PPG is present on most commercial smartwatches and smartbands. However, PPG is usually restricted to estimating the heart rate in these devices. On the other hand, PPG is being used as an important tool for the non-invasive estimation of several parameters, such as oxygen saturation, blood pressure, and glucose and bilirubin levels [3], [7].

In this work, we collected videos of voluntary patients' fingertips using a smartphone camera, simulating a PPG device. We pre-processed the videos to represent the intensity of the RGB channels in a three-dimensional time series. This dataset will be available for further experimentation. Since this data is complex, unstructured, and potentially noisy, we consider using deep artificial neural networks to estimate hemoglobin levels, obtained from the patients' blood count exams. However, deep networks can lead to other potentially critical problems in mHealth, such as the need for a large volume of labeled data for training, which is difficult to obtain.

With the original dataset containing only 106 examples, techniques to deal with small data may be necessary. Moreover, the data collected by smartphones may suffer from different types of noise, mainly because the patients may move their fingers during the gathering, and the presence of environmental light differences, among others. Considering this scenario, this paper evaluates the influence of different techniques on improving the model's generalization. These techniques are listed next.

We applied **varied neural architectures** following recommendations of the related literature. Moreover, to avoid the negative impacts of artifacts, we used **randomly selected subsequences** and assessed a simple **subsequences filter**. Finally, considering the small data scenario, we also experimented with **ensemble** and **data augmentation**.

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Combining these techniques, we performed a comprehensive experimental evaluation, analyzing nearly 150,000 predictions. Our results demonstrate how the assessed techniques improve the results compared to only using the dataset with straightforward preprocessing. Although our results demonstrate improvements in prediction, this task is challenging and has room for improvement. Therefore, we make the dataset available, so the community can freely use it to develop new research on this topic[1].

II. Background and Related Work

There are several studies that aim to estimate clinical parameters from PPG. This type of physiological signal has the potential to estimate heart rate, oxygen saturation, arterial pressure, blood hypoperfusion, peripheral vascular diseases, and other parameters [7], [8]. In particular, the level of hemoglobin is of great interest to the related literature [9]. In some cases, models based on simple statistics of the signal, such as wavelength, can achieve good correlations between the estimated parameter and the observed. Some other studies seek to improve the estimation of hemoglobin, among other parameters, using artificial intelligence techniques, such as a stack of regressors [10].

However, it should be noted that these articles use data obtained in a highly controlled manner. It is common for work proposing non-invasive hemoglobin estimation to focus mainly on the description of the hardware used or proposed for collection. Generally, the hardware used prioritizes external light control and better finger fixation as project priorities [11].

Nevertheless, there are studies using smartphones as the main data obtaining tool. For example, [11] used was collected by a modified smartphone containing a box to prevent external light interference and a dedicated LED array. In this case, as in many other works, the authors still rely on modifications, such as chambers to control the environment light.

There are cases where smartphones are used without modification. Even in these cases, the authors consider that there should be greater control over the environment, such as the use of lights with defined specifications used near the collection point [12], [13]. Usually, in these cases, the results achieved are lower than those obtained with devices modified for the task. However, there is a lack of efforts on deep learning for this task [9]. Therefore, we rely on the literature of deep learning for regression on time series data to develop this work.

A. Deep Learning for Time Series Extrinsic Regression

This work uses three deep neural network architectures, MLP, FCN, and ResNet [14]. The priority in selecting these architectures was to choose simpler models that could achieve high performance for the task without compromising the performance that deep neural networks offer. As originally proposed, the features such as the number of layers, neurons, kernel, and other model properties were kept, with the only change being the output which was no longer a softmax activation used in classification tasks, but rather a linear layer commonly used for regression.

B. Time Series Augmentation

The basic idea of data augmentation is to generate synthetic examples covering unexplored input space while maintaining correct labels. Therefore, it is necessary that the distortions applied to the data are neither too significant nor too soft. In the time series domain, data augmentation techniques are designed according to what the time series literature observed to be common distortions in that kind of data [15]. Some examples are the small variations of phase and trend, among others. However, these distortions are more or less adequate depending on the application domain.

As observed in [16], the transformation of temporal data can rely on signal processing techniques. Thus, distortions on the data might be applied in the time domain (such as variations in amplitude, cuts, and variations in *warping*), in the frequency domain (such as filters based on the Fourier transform) or even in the time-frequency domain.

III. Data Collection and Preprocessing

Photoplethysmography (PPG) consists of an optical measurement based on a light source (usually a light-emitting diode – LED) aimed at the skin. That light interacts with skin pigments, bones, muscles, and blood components, causing different levels of reflection or transmission captured by a photodetector, usually a photodiode. As explained in [6], obtaining a PPG from a smartphone follows a similar procedure. In this case, the camera performs the role of the photodetector, while the light source is the flash[2].

Each color channel of each frame of these videos is converted to a single value through the sum of all the pixel intensities. The time series from these values across the frames are standardized between 0 and 1 and receive an annotation of hemoglobin using the value (invasively) obtained through blood tests.

Due to the difficulty of gathering data, our dataset comprises PPG collected from only 63 patients, who were required to record 30 seconds of the index fingertip of each hand with a short interval between the recordings. Since some patients refused to collect data twice or aborted the recording before achieving 30 seconds, we ended with 106 signals.

IV. Experimental Evaluation

To evaluate the discussed techniques, we used 10-fold cross-validation. All the techniques applied in this work were performed separately in training and validation folds. Moreover, we repeat all the experiments five times to avoid misinterpretations of results related to stochastic factors, such as network initialization.

[1]All the source codes and the raw dataset, accompanied by a script to preprocess it, are freely available in a Git repository: https://github.com/diegofurts/mhealth_cbms2023

[2]Preliminary data analysis pointed out that keeping the flash on improves the signal quality. However, keeping the flash on for too long at the patient's fingertip may cause discomfort due to the high temperature. Therefore, we limited the videos to 30 seconds.

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To perform the experiments, we relied on *tsai*[3], an open-source library for time series Machine Learning. Using 200 epochs, we trained MLP, FCN, and ResNet. Besides, we assessed combining these architectures by the mean value in an ensemble-based model. Each of these models was used in seven different data manipulation scenarios, detailed in the following sections.

A. Random Windows

In all assessed scenarios, the first step of data manipulation is splitting the time series into five randomly chosen windows, comprising ten seconds of recording each. We hypothesize that short windows may improve the results mainly because they avoid noisy segments.

The predicted value for hemoglobin level may vary between the five windows. On the other hand, if the model sub or super estimates the hemoglobin in some subsequences, the median of the predicted values may represent a reasonable estimation. Following these assumptions, we use random windows in training and validation examples to create the `window` and `w+aug` scenarios. The second scenario adds data augmentation to the first one (c.f. Section IV-C).

B. Subsequence selection

As some windows include noisy parts, an optimal selection of one of the five segments may lead to a noisy-free dataset. We used the medoid to estimate the best subsequence from `window`, defined as the subsequence with the smallest average Euclidean distance to others. We also experiment with the three most significative segments according to this criterion.

In the scenarios that use the medoids (`medoid`, `3medoids`, `m-aug`, and `3m-aug`), we do not adopt the aggregation by median as the final prediction. Instead, we select the medoid of the randomly selected window from that example and predict a hemoglobin level to it.

C. Data Augmentation

In our experiments, we applied three types of distortions in the time domain using the *tsaug*, a Python package for time series augmentation[4]: "AddNoise", "TimeWarp", "Dropout".

We apply 19 distortions on each segment available for training. For example, as the `window` scenario transforms the signal into five segments, the total of training instances for that recording is 100 (19 augmentation plus the original data per window). Similarly, in the `medoid` scenario, each signal derives 20 examples in the training data, and 60 examples represent each segment in the `3medoids` scenario.

V. Results

The first experiment assesses the classifiers built on the full signals (scenario referred to as `full`), which are the baselines for hemoglobin estimation. Moreover, we also present the results of combining the three learning algorithms with our ensemble strategy. Surprisingly, MLP performed better, on average, than other methods, including the ensemble. It is a positive result once it is a simple, relatively low-cost, well-known architecture.

However, we note that `full` is the only scenario where MLP achieved the best results. In the scenarios with data pre-processing, we can observe reduced errors in all models, highlighting a more significant decrease for FCN, ResNet, and ensemble. The MLP achieved the worst average result in most scenarios. Table I presents all the results.

We can observe a few patterns in the results. Firstly, with two exceptions (MLP in `medoid` and `3medoids` scenarios), applying any of the techniques improves the results of all the models. Moreover, augmenting the data causes a positive impact in all cases. Finally, the `w+aug` is the scenario where all the models obtained the best results.

VI. Classification Analysis

Although we focus on estimating hemoglobin as a regression, the standard practice in the literature, this may hide some helpful information. For instance, consider a woman with a hemoglobin level of 11.5 g/dL, representing a low level, which may indicate different abnormalities, such as malnutrition and anemia. An absolute error of 0.7 indicates 10.8 g/dL (low level) or 12.2 g/dL (normal level). In other words, the same absolute error may represent a correct or an incorrect outcome.

For this reason, we also assessed converting our estimates into a binary class label for each time series (normal or abnormal). For male patients, we use the values from 13.8 to 17.2 g/dL as the normality range. For females, this range starts at 12.1 and goes to 15.1 g/dL. Table II presents the accuracy and f1 scores obtained in this experiment.

The augmentation, windowing, and cleaning techniques consistently improved the accuracy. On the other hand, the best f-1 scores were commonly obtained in the `full` scenario. That indicates that all the applied techniques usually benefit the majority class, which, in this case, is the abnormality. There are two exceptions for this claim: MLP and ensemble. In all the cases, MLP obtained an f-1 score slightly better using any technique compared to the `full` scenario. We observe similar behavior for the ensemble in augmented data cases.

VII. Discussion and Final Remarks

One of the contributions of this paper is a novel dataset for real-world smartphone-based hemoglobin level prediction. Similar datasets found in the literature usually rely on more complex setups, including chambers or other mechanisms to avoid light differences and unexpected movements by the patient. To obtain the first results for this dataset, we focused on the assumptions and questions presented in the introductory section. Therefore, we measured the impact of different techniques in dealing with small and noisy data.

Such techniques usually benefit the results but with some differences between error and its parallel in classification. For instance, randomly selecting five smaller signal windows and applying data augmentations (`w+aug`) consistently achieved the best results regarding the absolute error. On the other hand,

[3] https://timeseriesai.github.io/tsai/
[4] https://tsaug.readthedocs.io

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TABLE I

ABSOLUTE ERRORS (WITH STANDARD DEVIATION WITHIN THE PARENTHESES) OF HEMOGLOBIN LEVEL PREDICTION. THE UNDERLINED RESULTS REPRESENT THOSE WHO OVERCOME THE `FULL` SCENARIO USING THE SAME ARCHITECTURE. THE **BOLD** CELLS REPRESENT THE BEST RESULT FOR EACH ARCHITECTURE. NO RESULTS OBTAINED IN THE `FULL` SCENARIO ARE UNDERLINED OR BOLD.

	`full`	`window`	`w+aug`	`medoid`	`m+aug`	`3medoids`	`3m-aug`	Best
MLP	1.75 (1.27)	1.60 (1.22)	**1.57 (1.17)**	1.80 (1.79)	1.63 (1.27)	1.80 (2.68)	1.63 (1.24)	1.57 (1.17)
FCN	2.24 (1.65)	1.54 (1.14)	**1.54 (1.13)**	1.71 (1.39)	1.61 (1.23)	1.71 (1.20)	1.61 (1.22)	1.54 (1.13)
ResNet	1.95 (1.52)	1.55 (1.15)	**1.52 (1.14)**	1.84 (1.57)	1.55 (1.21)	1.84 (1.35)	1.55 (1.20)	1.52 (1.14)
Ensemble	1.83 (1.36)	1.55 (1.14)	**1.53 (1.13)**	1.63 (1.33)	1.54 (1.20)	1.63 (1.30)	1.54 (1.17)	1.53 (1.13)
Best	1.75 (1.27)	1.54 (1.14)	1.52 (1.14)	1.63 (1.33)	0.73	1.63 (1.30)	1.54 (1.17)	

TABLE II

RESULTS OF THE CLASSIFICATION BETWEEN NORMAL AND ABNORMAL HEMOGLOBIN LEVELS. THE UNDERLINED RESULTS REPRESENT THOSE WHO OVERCOME THE `FULL` SCENARIO USING THE SAME ARCHITECTURE. THE **BOLD** NUMBERS REPRESENT THE BEST RESULT FOR EACH ARCHITECTURE. NO RESULTS OBTAINED IN THE `FULL` SCENARIO ARE UNDERLINED OR BOLD.

	`full`		`window`		`w+aug`		`medoid`		`m+aug`		`3medoids`		`3m+aug`		Best	
	acc	f1	acc	f1	acc	f1	acc	f1	acc	f1	acc	f1	acc	f1	acc	f1
MLP	0.64	0.52	0.68	0.54	0.66	0.53	**0.70**	0.54	0.69	0.53	0.68	**0.54**	0.67	0.51	0.70	0.54
FCN	0.68	0.51	0.69	0.46	0.69	0.45	0.66	0.53	**0.73**	0.45	0.71	0.44	0.72	0.44	0.73	0.51
ResNet	0.65	0.54	0.68	0.49	0.67	0.52	**0.72**	0.52	0.69	0.51	0.70	0.50	0.70	0.53	0.72	0.54
Ensemble	0.66	0.49	0.68	**0.51**	0.67	0.48	0.70	**0.51**	**0.72**	0.49	0.68	0.49	0.71	0.50	0.72	0.51
Best	0.68	0.54	0.69	0.54	0.69	0.53	0.72	0.54	0.73	0.53	0.71	0.54	0.72	0.53		

filtering these subsequences before augmenting is better in the classification scenario.

The most remarkable similarity between regression and classification is that augmenting the dataset consistently improves the results. Therefore, our primary recommendation is to employ data augmentation. We intend to investigate the impact of other augmentations in the task, including distortions in the frequency and time-frequency domains.

Finally, the data acquisition showcased in this study is very promising as it correlates with the measured parameter when coupled with machine learning techniques without requiring specialized hardware or specific ambient light control. Hence, we aim to create a larger dataset and investigate these approaches in different data volume scenarios.

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